

Johnny Clem

AI-Native Engineer with Pre-AI Principles

Austin, TX · @johnnyclem on GitHub, LinkedIn, and X · smallchat.dev

SUMMARY

Systems-minded engineer with 15+ years shipping developer tools, real-time media frameworks, and agentic AI systems across iOS, macOS, and server platforms. Author of smallchat, an open-source TypeScript semantic tool dispatch compiler built on MCP-adjacent patterns. Built and shipped JCAvascript, a production MCP server enabling AI agents to drive macOS automation through structured tool invocation. Deep experience translating complex platform capabilities into accessible developer experiences through code, documentation, and technical advocacy.

TECHNICAL SKILLS

Languages: TypeScript, Python, Swift, Objective-C, C/C++, Bash

AI / Agent Systems: MCP server design & implementation, tool dispatch/invoke, RAG pipelines, vector databases (sqlite-vec), ONNX embeddings, LLM integration (Anthropic, OpenAI APIs), prompt engineering, agent orchestration

Platforms & Infrastructure: macOS, iOS, Linux, Docker, CI/CD (CircleCI, GitHub Actions), SQLite, Node.js, npm package publishing

Developer Tools: Claude Code, GitHub Copilot, Cursor, Xcode, VS Code, Git

Advocacy: Technical blogging, conference presentations, video production, open-source maintainership

PROFESSIONAL EXPERIENCE

Creator & Maintainer | smallchat (Open Source) | 2024–Present

- Designed and built @smallchat/core, a TypeScript semantic tool dispatch compiler that treats LLM tool-call token bloat as a compilation problem, mapping Objective-C/Smalltalk runtime concepts onto modern agent architectures
- Implemented security hardening, worker thread isolation, SQLite persistence, and container sandboxing across multiple audit cycles
- Built vector index search for semantic tool matching, enabling efficient tool selection at scale without exhaustive schema enumeration
- Published to npm as @smallchat/core; developed companion marketing site (smallchat.dev) and Swift port (smallchat-swift)

Creator & Maintainer | JCAvascript (Open Source + MCP Server) | 2024–Present

- Built and maintain an open-source Swift package (johnnyclem/JCAvascript) for programmatic AppleScript execution with object and variable passing across the OSA bridge
- Designed and shipped a companion MCP server enabling AI assistants to invoke macOS automation through structured tool schemas—in production use for agentic workflows
- Defined tool invocation patterns and context exchange interfaces that allow LLM-driven agents to interact with native macOS services reliably

Staff iOS Engineer | Realtor.com (Move, Inc.) | 2024–Present

- Drive architecture and code quality on a large-scale Swift/UIKit iOS application serving millions of users
- Optimized CI/CD pipeline performance, porting DerivedData caching strategies to reduce build times
- Collaborate across distributed engineering teams on shared infrastructure and notification systems

Interim CTO | Wire Network / AI Layer Labs | 2023–2024

- Led development of an agentic macOS application with agent memory, multi-model tool use, and native OS integration via AppleScript bridge
- Produced technical video podcast documenting agentic development workflows (~100K downloads/week at peak)
- Designed agent orchestration patterns and MCP-style tool integration for cross-platform AI assistants

Senior Software Engineer | Truepic | 2022–2023

- Built authenticated media capture pipelines leveraging device-level attestation and secure enclaves

CTO & Co-Founder | Cinegenix (Filmic Pro) | 2012–2020

- Led engineering for the #1 professional camera app on iOS, used by filmmakers on productions for Apple, Netflix, and Sundance selections
- Architected real-time video pipeline on AVFoundation supporting LOG/flat gamma, manual exposure, focus peaking, and live color grading
- Built and managed developer team; defined technical roadmap translating filmmaker needs into shipping software
- Created extensive technical documentation, tutorials, and developer resources adopted across the mobile filmmaking community

NOTABLE PROJECTS

- ContextBench — Python LLM benchmark extending HumanEval/SWE-bench with remaining-context-window testing dimension; async runner, Anthropic/OpenAI provider abstraction, degradation scoring
- MidiClaw — Native macOS MIDI agent using on-device LLMs with custom ~3M parameter transformer adapter; MLX backend, predictive scheduling, feedback loop prevention
- birdtalk — iOS field sampler app with on-device ML species classification (Google Perch), real-time audio DSP via AVAudioEngine and Accelerate framework
- Polytician — Open-source MCP server for shared AI agent memory; pure TypeScript with ONNX embeddings and sqlite-vec for vector-based semantic retrieval

EDUCATION

Certificate Program, Applied AI & Machine Learning — University of Washington

B.A., Music Theory — Northwest University

A.A., General Studies — Green River Community College